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Batch : J2

Assignment - 1

- **Title**

Design a network : (use packet tracer / GNS).
Design a simple network (LAN) with different topologies & test it using PING utility.

- **Objective**

1. To learn & understand the concept of LAN & test it using PING command.
2. To design a network using various network topologies in cisco packet tracer.

- **Theory**

A local area network (LAN) is a collection of devices connected together in one physical location, such as a building, office, or home. A LAN can be small or large, ranging from a home network with 2 users to an enterprise network with 1000s of users & devices in a office / school. A LAN comprises cable, access points, switches, routers & other components that enable devices to connect to internal servers, web servers & other LANs via wide area networks.

The advantage of a LAN are the same as those for any group of devices networking together. The devices can use a single internet connection, share files with one another, print to shared printers, & be accessed & even controlled by one another.

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In general there are two types of LANs: client-server LANs & peer-to-peer LANs. A client/server LAN consists of several devices (the client) connected to a central server. The server manages file storage, application access, device access & network traffic. A client can be any connected device that runs or accesses application or the internet. The clients connect to the server either with cables (b) wireless connections.

A peer-to-peer LAN doesn't have a central server & can't handle heavy workloads like client-server LAN can, & so they're typically smaller. On a peer-to-peer LAN, each device share equally in the functioning of the network.

- Network topologies

It is arrangement with which computer systems network devices are connected to each other.

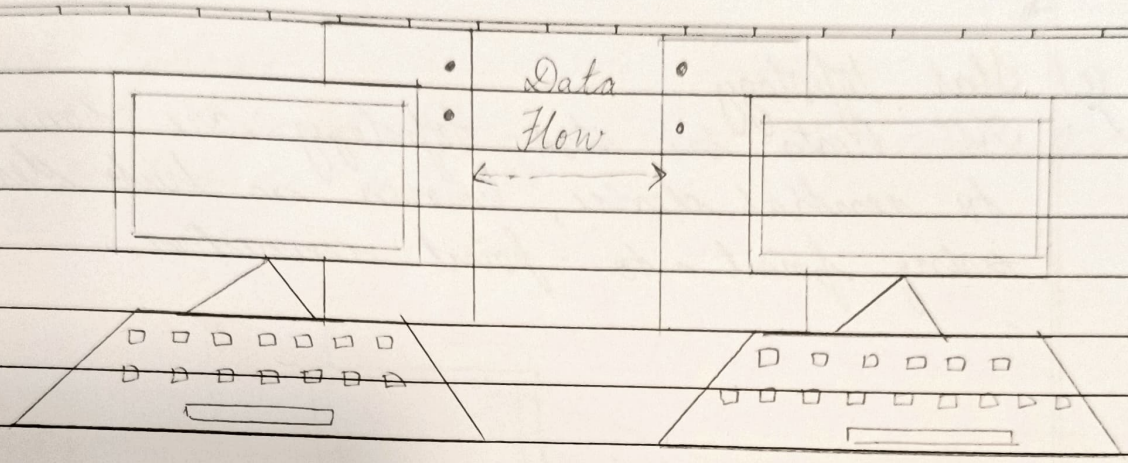
Types of topologies

(a) Point-to-point

Point-to-point networks contains exactly two such as computers, switches or routers, servers connected back to back using a single piece cable.

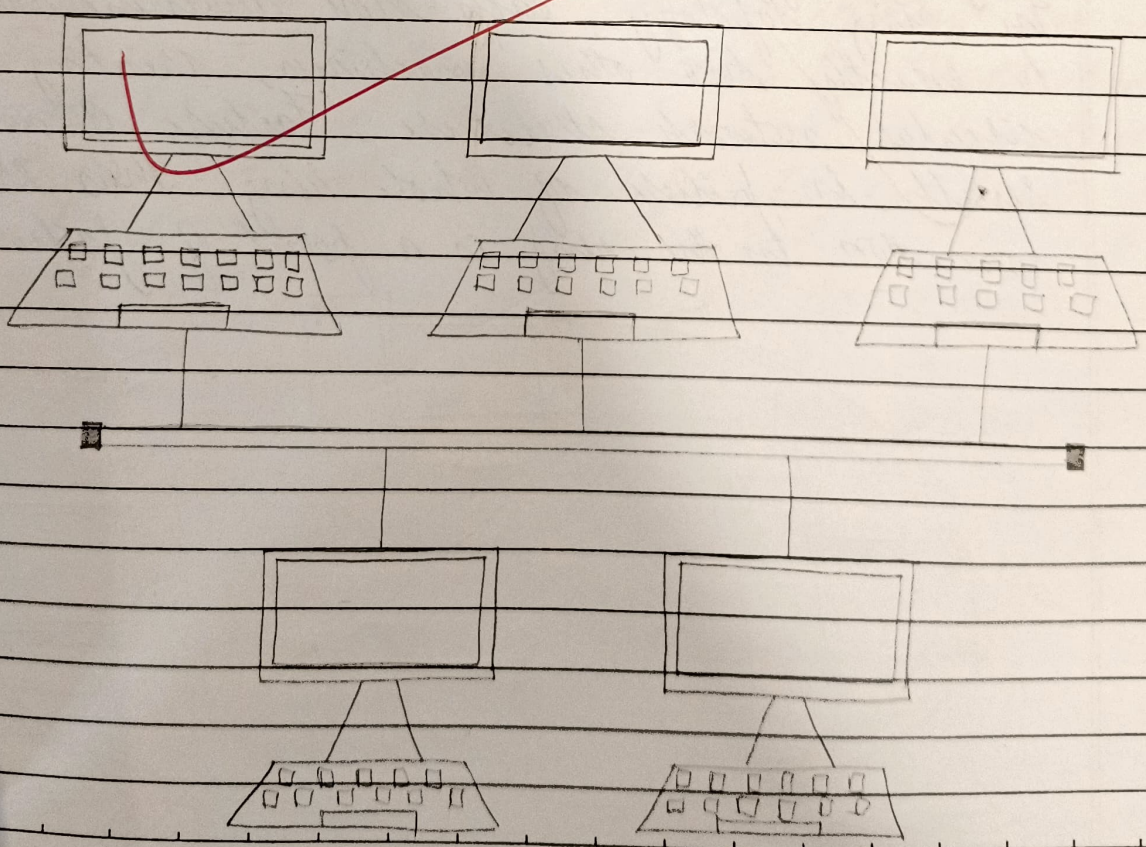
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(b) Bus topology

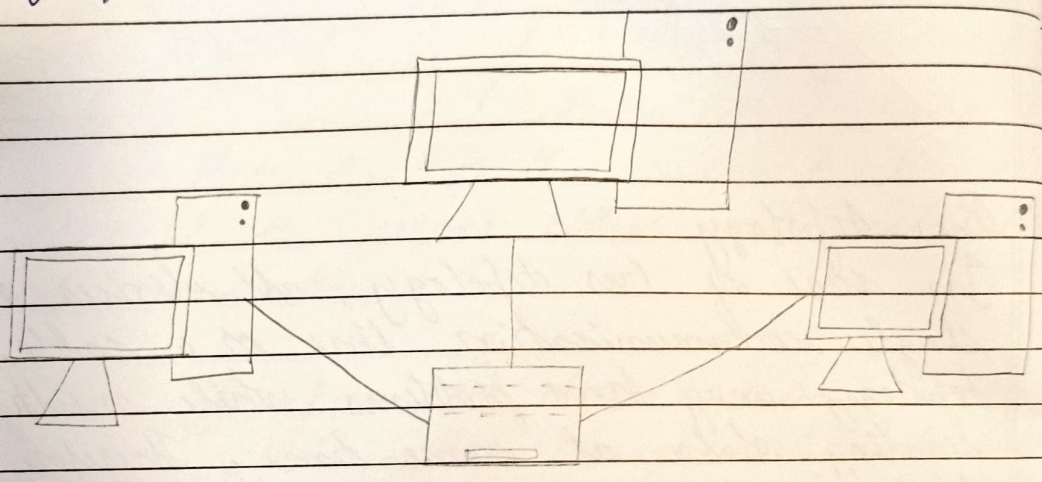
In case of bus topology, all devices share a single communication line or a cable. Bus topology may have problems while multiple hosts sending data at same time. Therefore, bus topology either uses CMD/CD technology or recognises one host as bus master to solve the issue.



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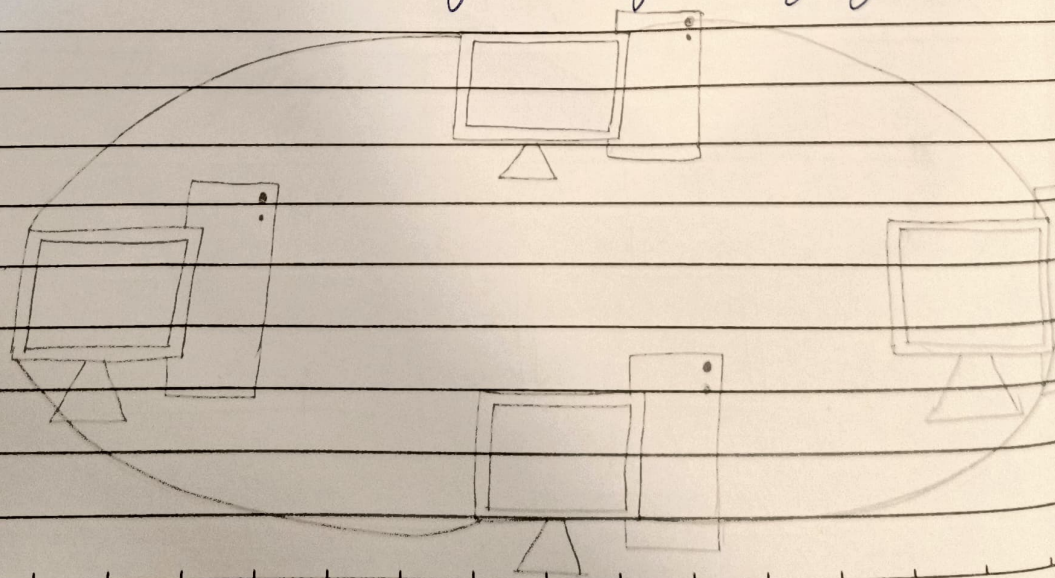
(c) Star topology.

All hosts in star topology are connected to central device, known as hub device, using point-to-point connection.



(d) Ring topology.

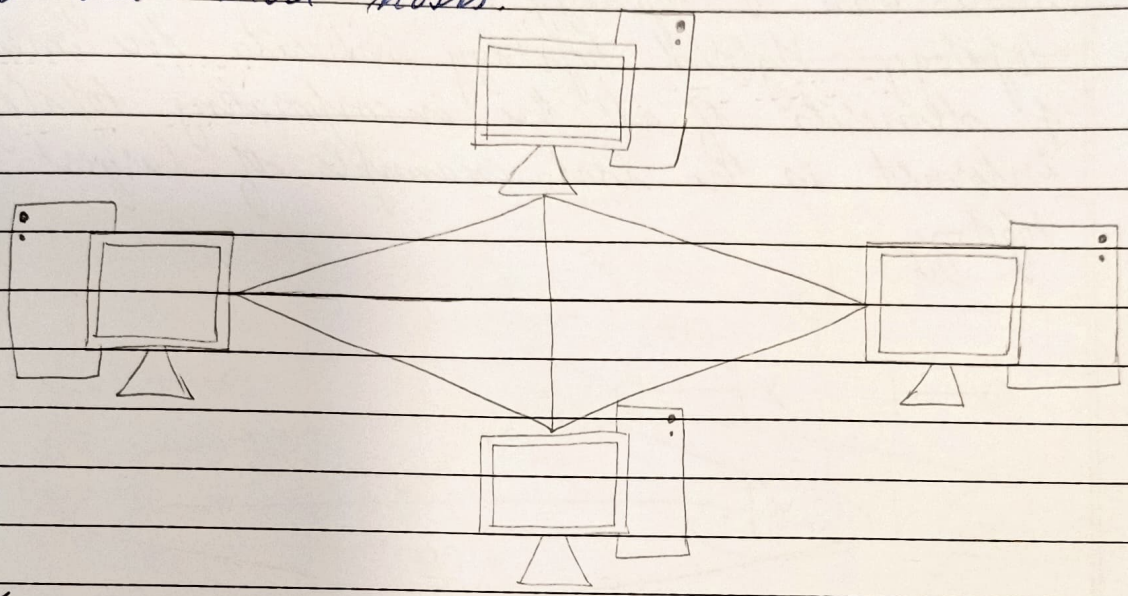
In ring topology, each host machine connected to exactly two other machines, creating a circular network structure. Failure of any one results in failure of whole ring. Thus every connection in the ring is a point of failure.



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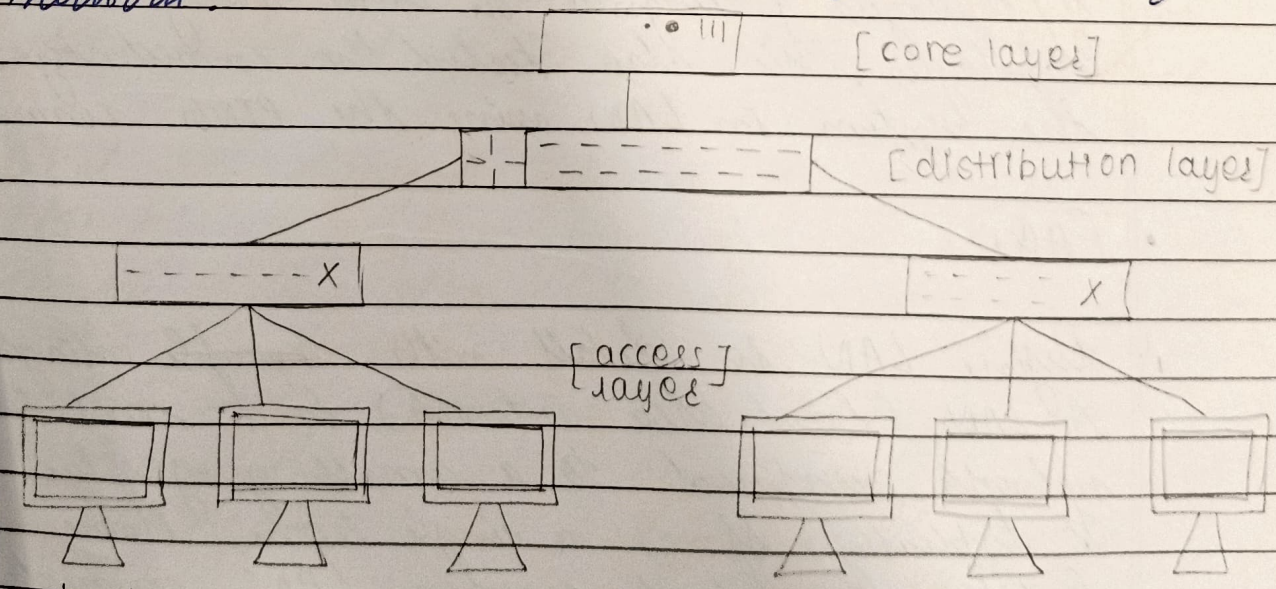
(e) Mesh topology

In this type of topology, host is connected to one / more hosts.



(f) Tree topology

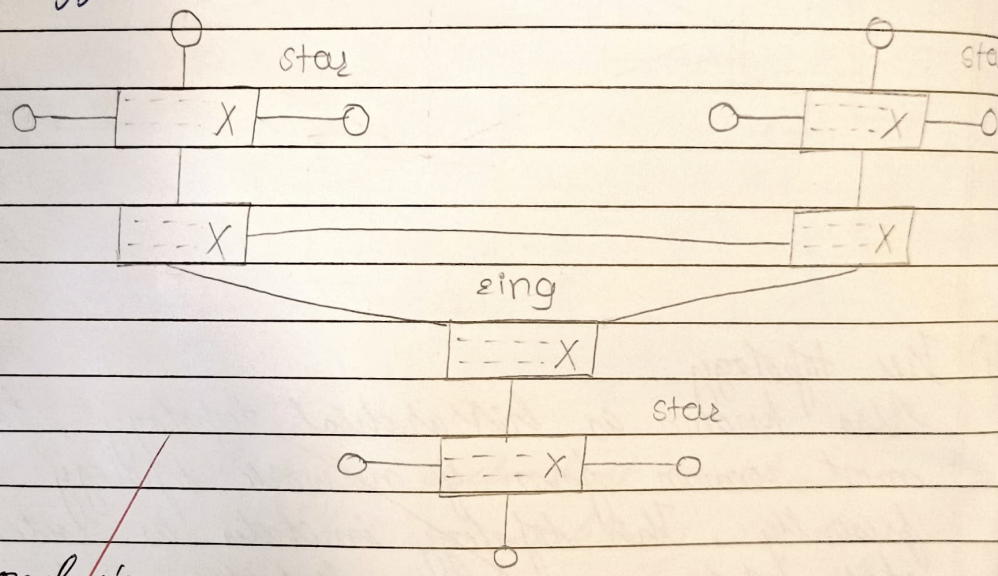
Also known as hierarchical topology, this is most common form of network topology in use presently. This topology imitates as extended star topology & inherits properties of bus topology & divides network into multiple layers / levels of network.



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(g) Hybrid topology

A network structure whose design contains more than 1 topology is said to be hybrid topology. Hybrid topology inherits the merits & demerits of all the incorporating topologies. internet is the best example of largest hyb topology.



- Conclusion

Thus we studied various types of networks, ty of topologies & designed an LAN using various topologies. We also tested the connectivity bt the system in LAN using the PING comman

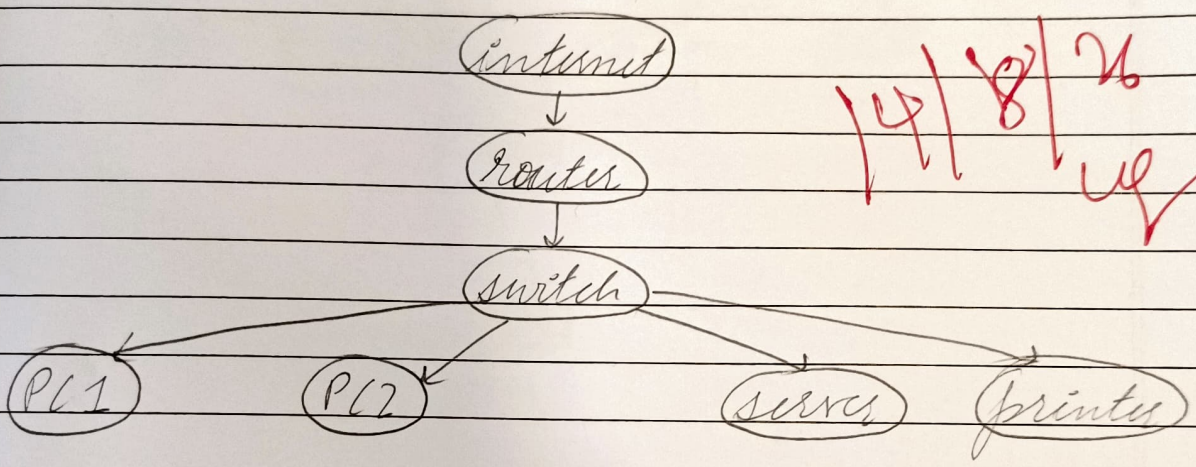
- FAQs

1) Define LAN in detail with ~~example~~ diagra

A LAN (local area network) is a computer network confined to a small geographic a It typically spans a single room, home o building or school campus. LAN connects

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local workstations, personal computers, printers, & servers to share resources & exchange data efficiently. They offer high data transfer rates & low propagation delays with lower error rates.



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24 Explain PING utility.

PING (packet internet groper) is a fundamental network diagnostic tool used to test the reachability of a host on an internet protocol (IP) network. It confirms if a specific server/device is online & active. It also calculates the round trip time, which is the time taken for data to travel to the destination and back.

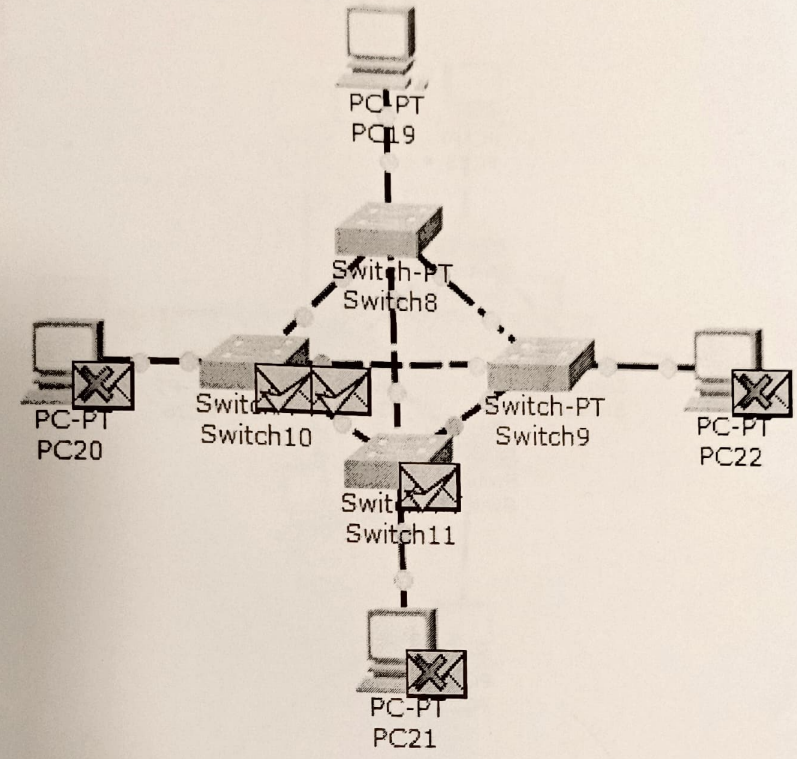
31 What are different types of LAN topologies?

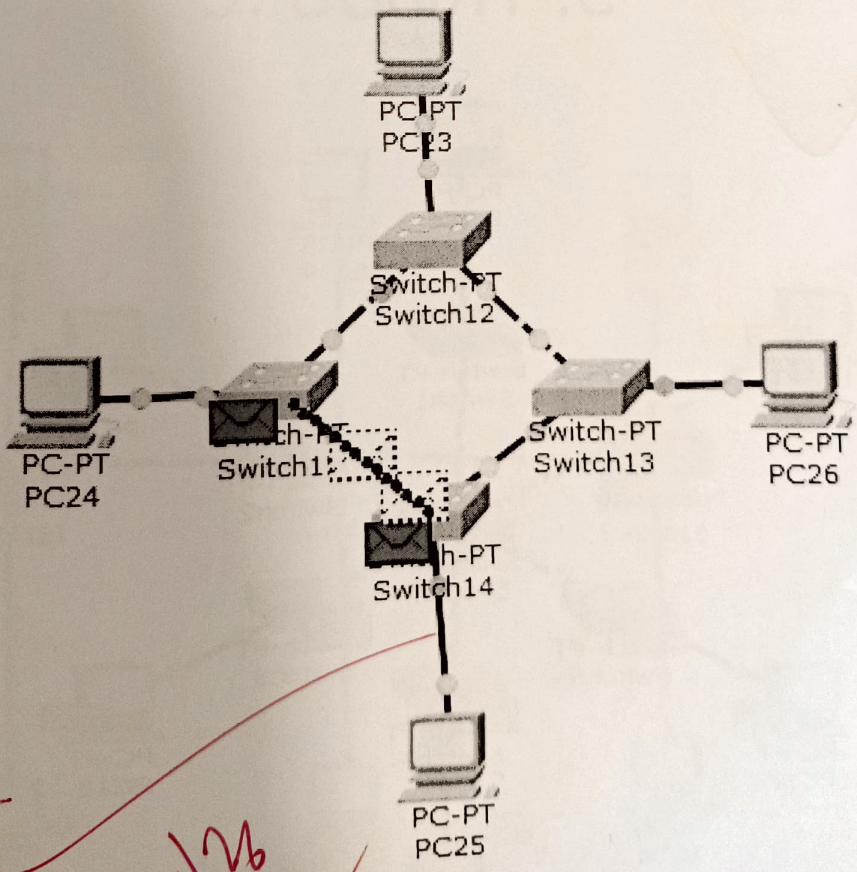
LAN topology defines how different devices & computers are structurally arranged & connected within a network.

- Bus topology
- Star topology
- Tree topology
- Ring topology
- Mesh topology
- Hybrid topology

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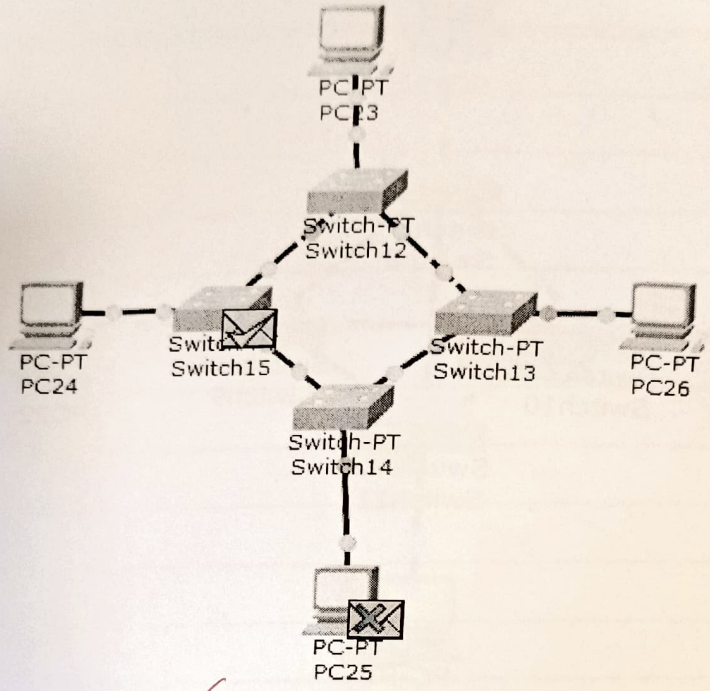
1. MESH


PC-PT
PC19
Switch-PT
Switch8
PC-PT
PC20
Switch10
Switch-PT
Switch9
PC-PT
PC22
Switch11
PC-PT
PC21

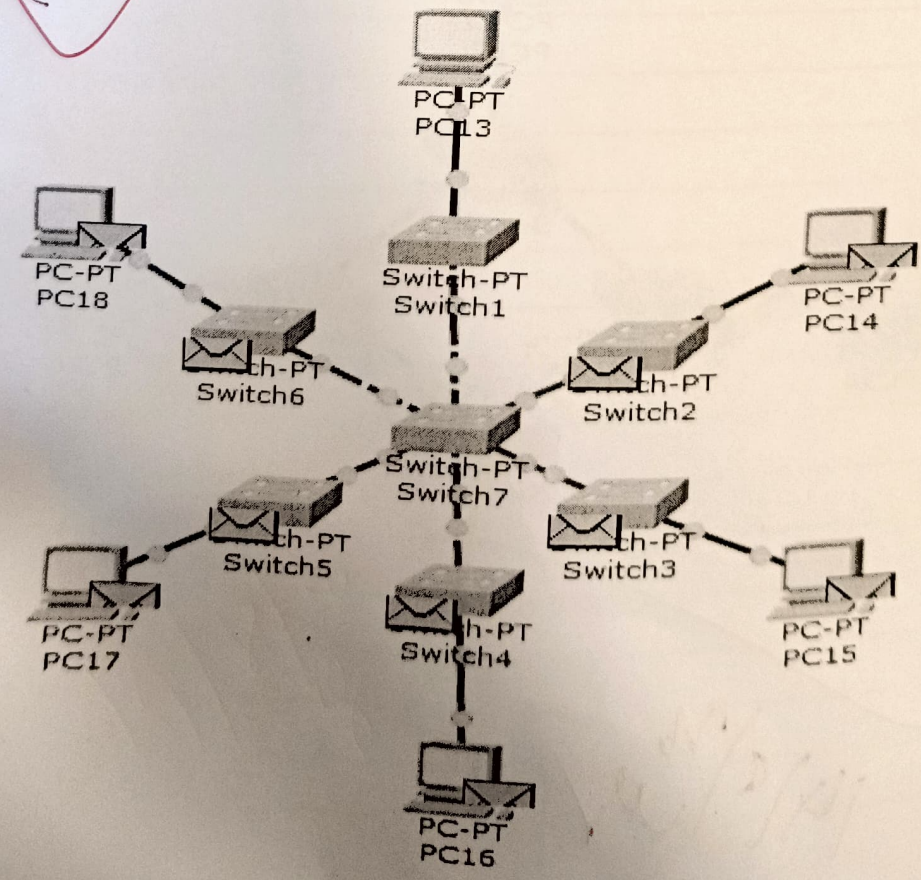

PC-PT
PC23
Switch-PT
Switch12
PC-PT
PC24
Switch13
PC-PT
PC26
Switch14
PC-PT
PC25

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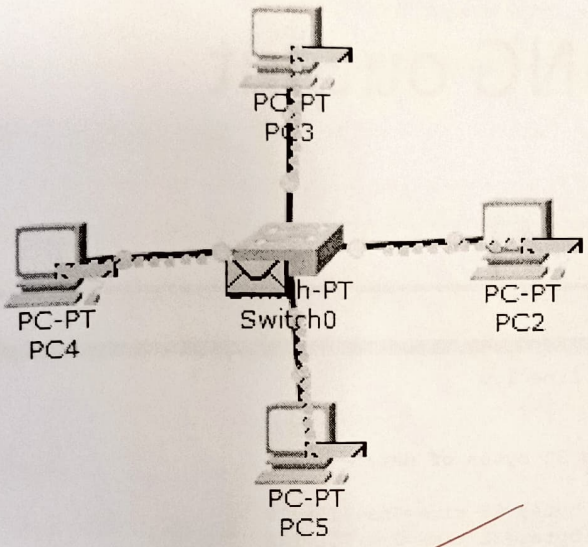
2. RING


PC-PT
PC13
Switch-PT
Switch12
PC-PT
PC24
Switch-PT
Switch15
Switch-PT
Switch13
PC-PT
PC26
Switch-PT
Switch14
PC-PT
PC25

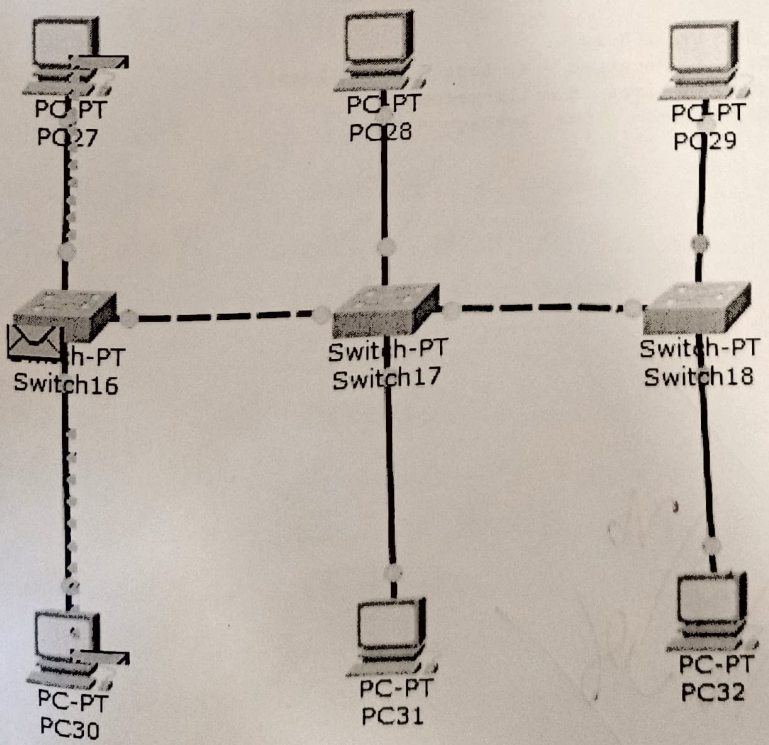
3. TREE


PC-PT
PC13
PC-PT
PC18
Switch-PT
Switch1
PC-PT
PC14
Switch-PT
Switch6
Switch-PT
Switch2
Switch-PT
Switch7
Switch-PT
Switch5
Switch-PT
Switch3
PC-PT
PC17
Switch-PT
Switch4
PC-PT
PC15
PC-PT
PC16

4. STAR


PC-PT
PC3
PC-PT
PC4
h-PT
Switch0
PC-PT
PC2
PC-PT
PC5

5. BUS


PC-PT
PC27
PC-PT
PC28
PC-PT
PC29
h-PT
Switch16
Switch-PT
Switch17
Switch-PT
Switch18
PC-PT
PC30
PC-PT
PC31
PC-PT
PC32

PING output

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time=7ms TTL=128
Reply from 192.168.10.4: bytes=32 time=2ms TTL=128
Reply from 192.168.10.4: bytes=32 time=0ms TTL=128
Reply from 192.168.10.4: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 3ms

PC>ping 192.168.10.5

Pinging 192.168.10.5 with 32 bytes of data:

Reply from 192.168.10.5: bytes=32 time=0ms TTL=128
Reply from 192.168.10.5: bytes=32 time=0ms TTL=128
Reply from 192.168.10.5: bytes=32 time=0ms TTL=128
Reply from 192.168.10.5: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.10.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

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